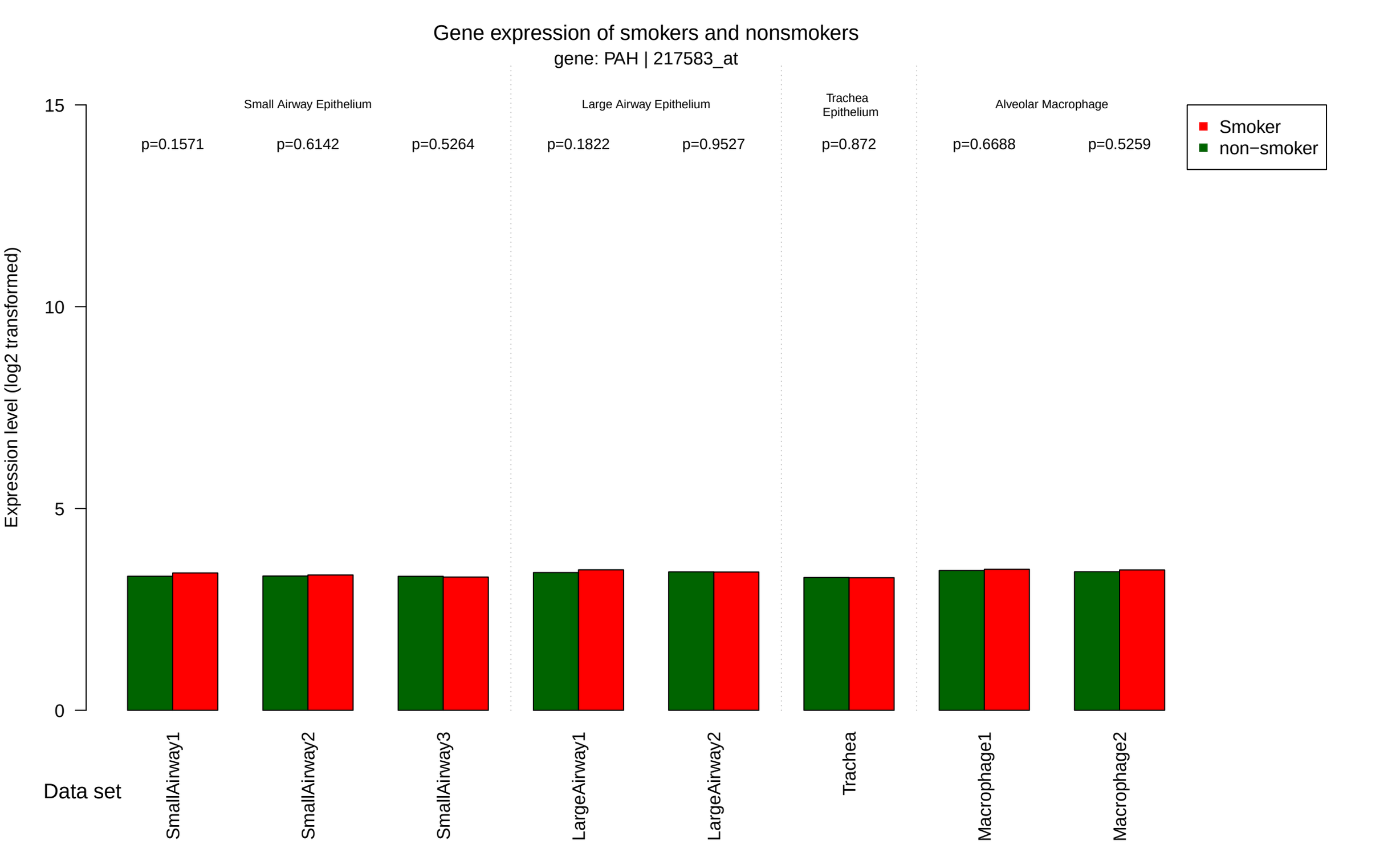
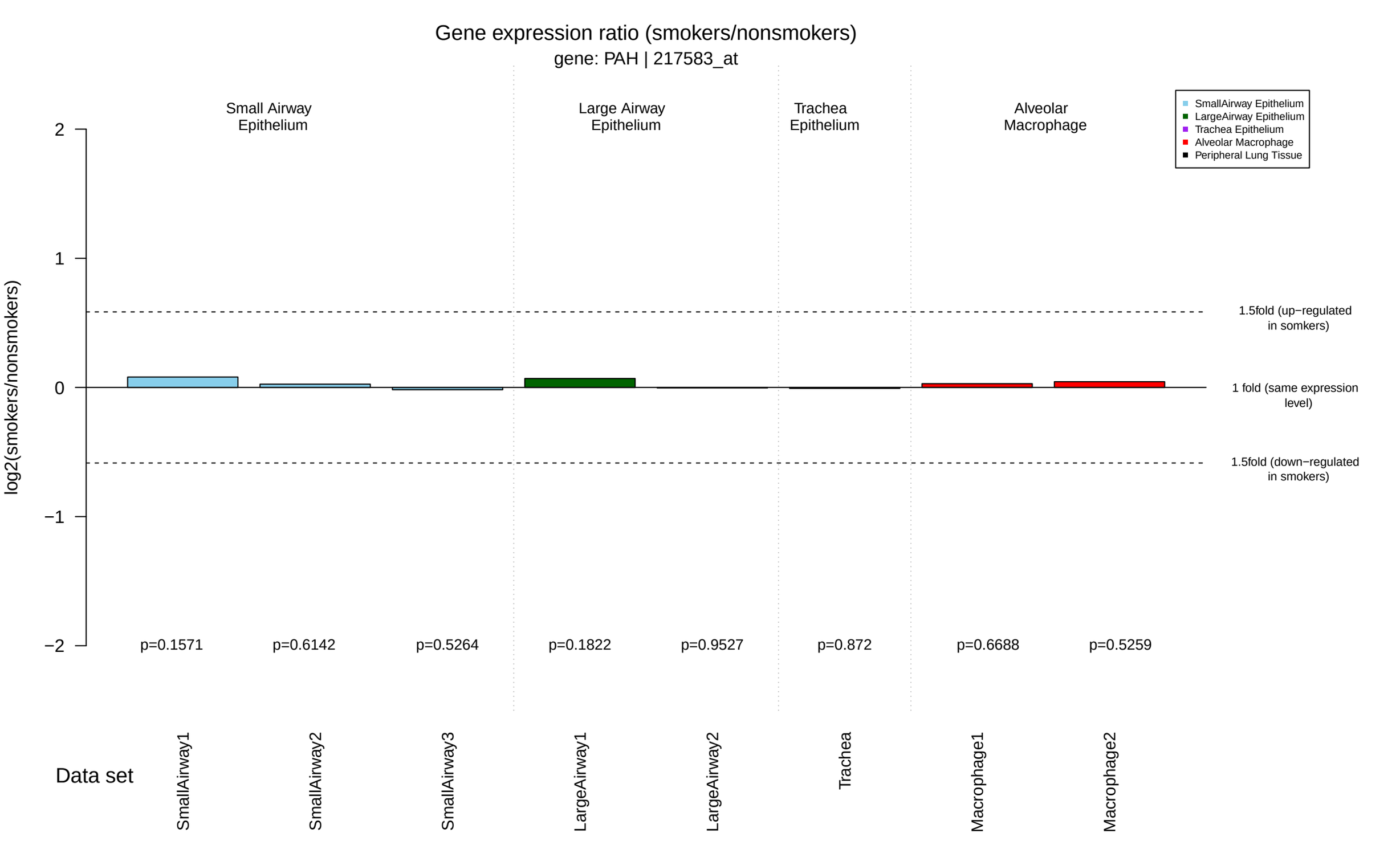
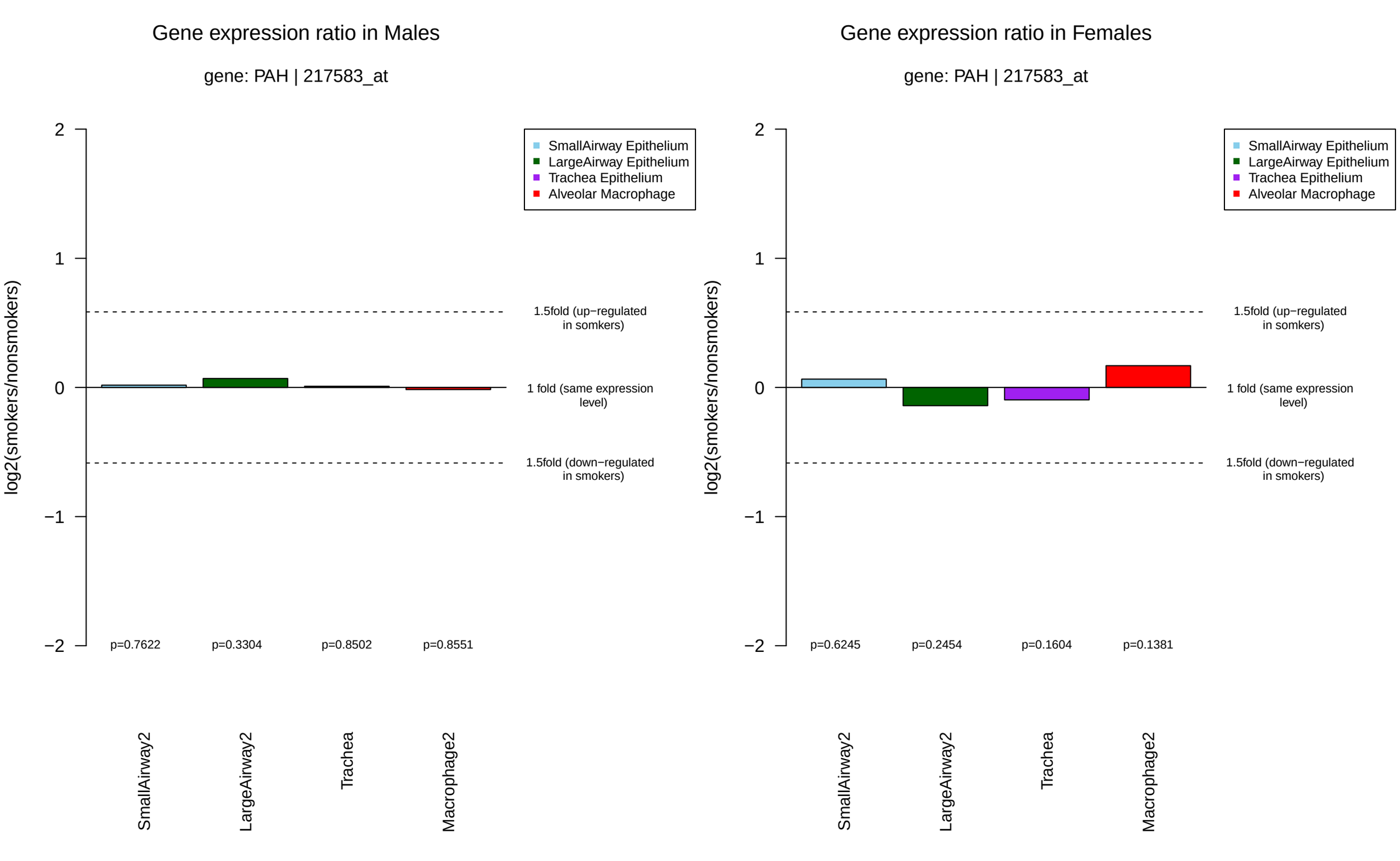
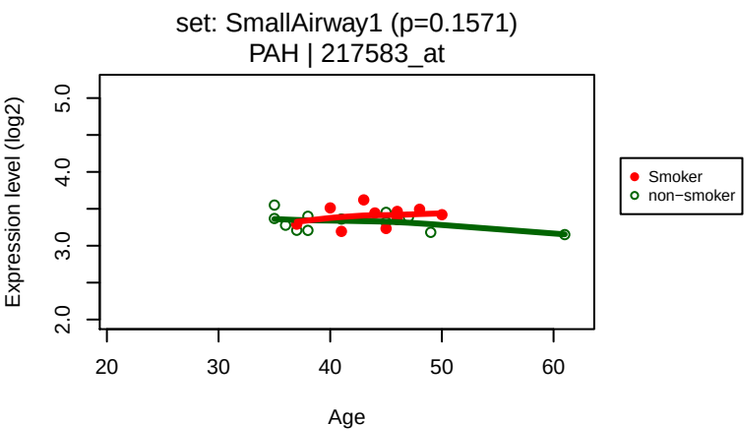
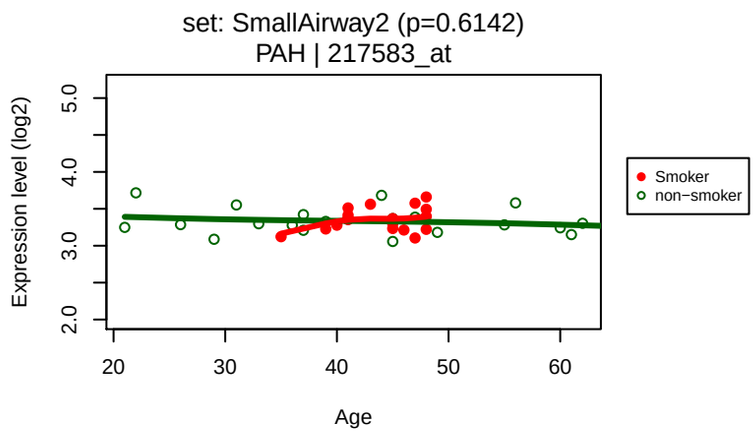
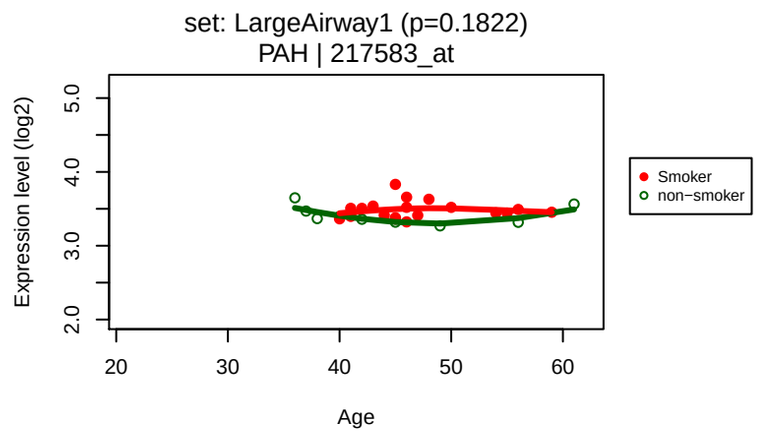

Gene expression of smokers and nonsmokers
gene: PAH | 217583_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1571
p=0.6142
p=0.5264
p=0.1822
p=0.9527
p=0.872
p=0.6688
p=0.5259
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

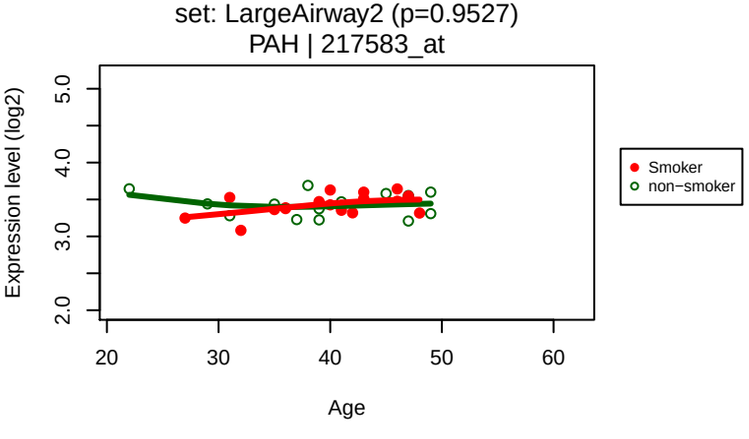

Gene expression ratio (smokers/nonsmokers)
gene: PAH | 217583_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1571
p=0.6142
p=0.5264
p=0.1822
p=0.9527
p=0.872
p=0.6688
p=0.5259
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

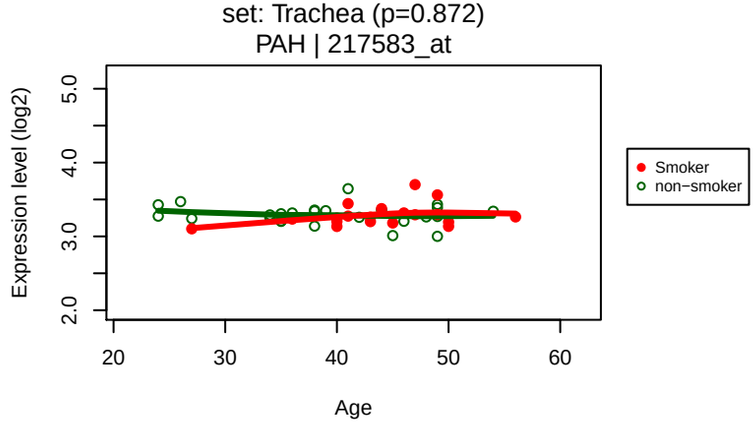

Gene expression ratio in Males
gene: PAH | 217583_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7622
p=0.3304
p=0.8502
p=0.8551
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAH | 217583_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6245
p=0.2454
p=0.1604
p=0.1381
SmallAirway2
LargeAirway2
Trachea
Macrophage2

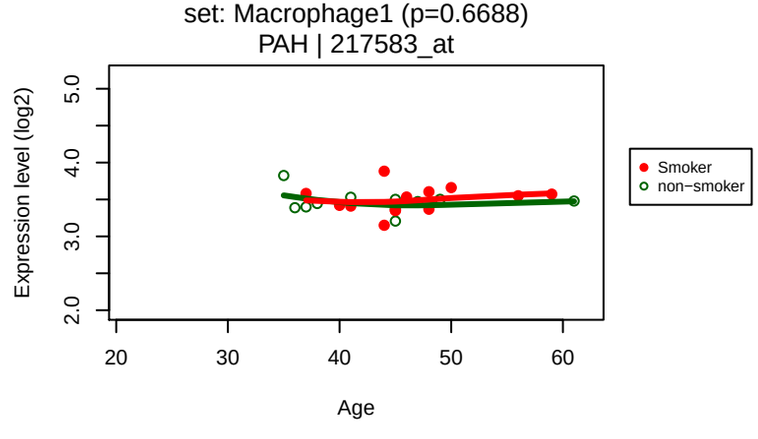

set: SmallAirway1 (p=0.1571)
PAH | 217583_at
Expression level (log2)
Age
Smoker
non−smoker

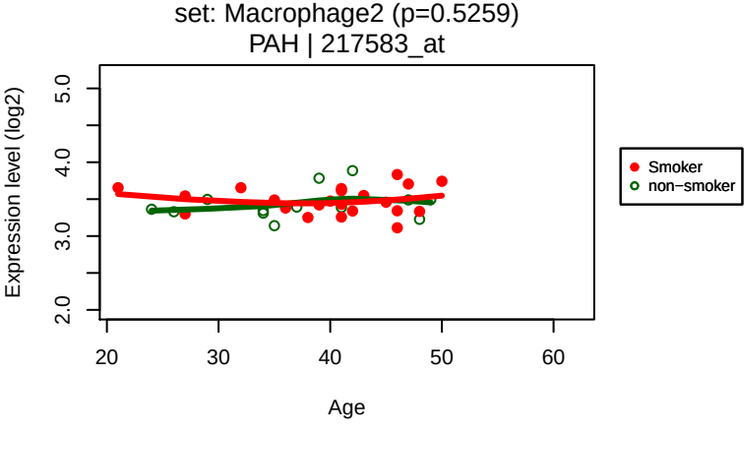

set: SmallAirway2 (p=0.6142)
PAH | 217583_at
Expression level (log2)
Age
Smoker
non−smoker

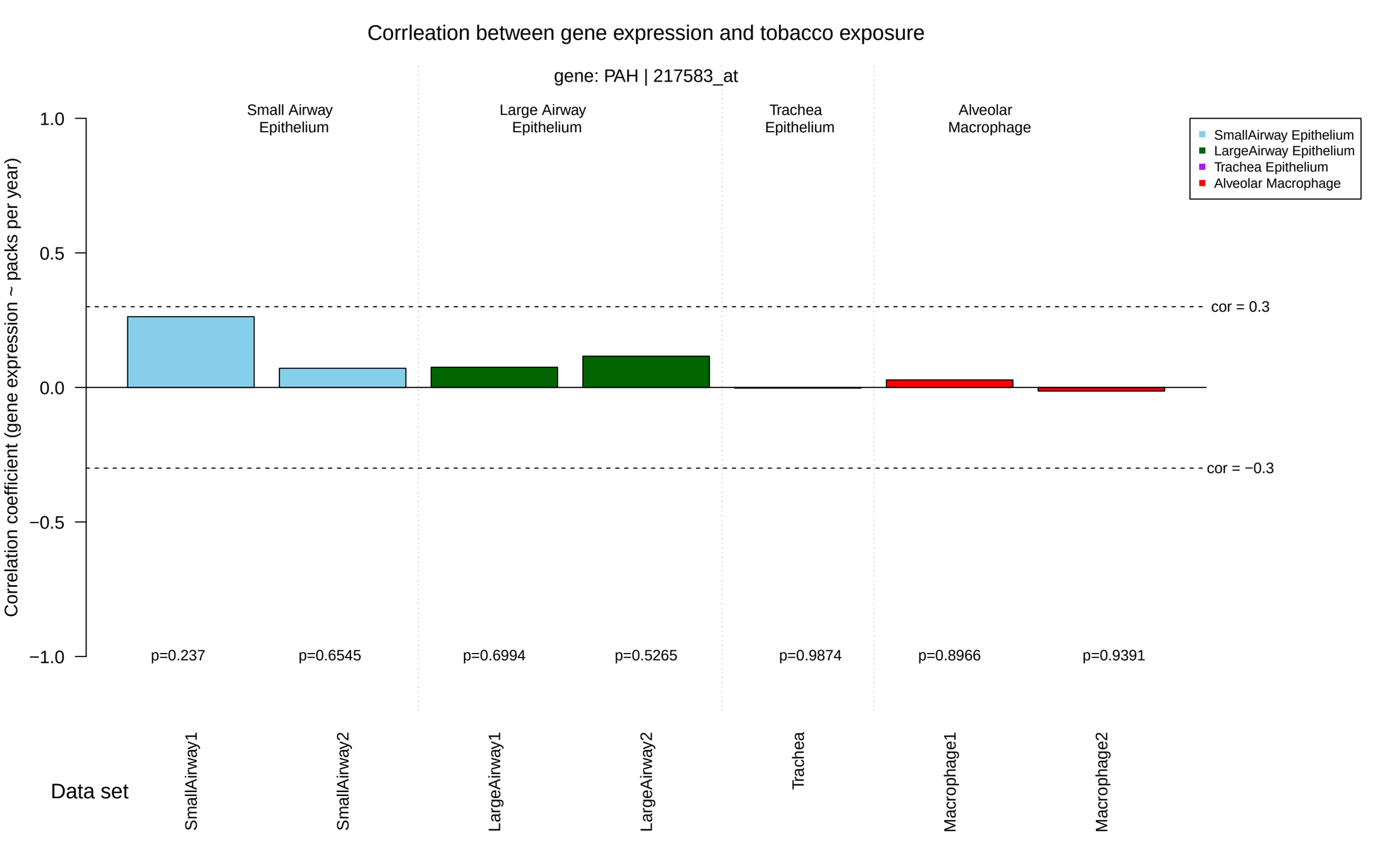

set: LargeAirway1 (p=0.1822)
PAH | 217583_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9527)
PAH | 217583_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.872)
PAH | 217583_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6688)
PAH | 217583_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5259)
PAH | 217583_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAH | 217583_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.237
p=0.6545
p=0.6994
p=0.5265
p=0.9874
p=0.8966
p=0.9391
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2